Saad Hossain

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Education

Bachelor of Applied Science (BASc), Biomedical Engineering Sep 2020 – Apr 2025
University of Waterloo *GPA: 94/100*

- Specialization in Medical Artificial Intelligence.
- Research supervisors: Sirisha Rambhatla, Paul Fieguth, Kerstin Dautenhahn, Katja Mombaur.

Publications

- [1] **S. Hossain**, T. Tseng, P. Pandey, S. Vajpayee, M. Kowal, N. Nonta, S. Simko, S. Casper, Z. Jin, K. Pelrine, S. Rambhatla. TamperBench: Systematically Stress-Testing LLM Safety Under Fine-Tuning and Tampering. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (KDD), 2026. [Link] [GitHub]
  ⋆ *Oral presentation (top 4.5% of papers).*

- [2] K. Kaai, **S. Hossain**, S. Rambhatla. Bringing Equity to Classification: Domain Generalization for Domain-Linked Classes. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (T-PAMI), 2025. [Link] [GitHub]

- [3] R. Chaudhary, J. Ho, D. Smith, **S. Hossain**, J. Hargun, B. VanBerlo, N. Murphy, R. Prager, K. Rikhraj, J. Tschirhart, R. Arntfield. Diagnostic Accuracy of an Automated Classifier for the Detection of Pleural Effusions in Patients Undergoing Lung Ultrasound. *American Journal of Emergency Medicine*, 2025. [Link]

- [4] **S. Hossain**, S. Vajpayee, S. Rambhatla. SafeTuneBed: A Safety Assessment Framework for Harmful Finetuning Defenses. *International Joint Conference on Artificial Intelligence* (IJCAI) Workshop on User-Aligned Assessment of Adaptive AI Systems, 2025. [Link] [GitHub]

- [5] C. Liu, **S. Hossain**, C. Thomas, K. Lai, R. Vemulapalli, S. Rambhatla, A. Wong. LangDA: Language-guided Domain Adaptive Semantic Segmentation. *Advances in Neural Information Processing Systems* (NeurIPS) Workshop on Adaptive Foundation Models, 2024. [Link]
  ⋆ *Best Vision Paper Award at CVIS 2024.*

- [6] J. Park, K. Kaai, **S. Hossain**, N. Sumi, S. Rambhatla, P. Fieguth. Domain-Guided Spatio-Temporal Self-Attention for Egocentric 3D Pose Estimation. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (KDD), 2023. [Link] [GitHub]

- [7] K. Kaai, **S. Hossain**, S. Rambhatla. Are All Classes Created Equal? Domain Generalization for Domain-Linked Classes. *Advances in Neural Information Processing Systems* (NeurIPS) Workshop on Distribution Shifts: New Frontiers with Foundation Models, 2023. [Link] [GitHub]

- [8] J. Park, F. Barnard, **S. Hossain**, S. Rambhatla. Implicit Stylization for Domain Adaptation. *International Conference on Learning Representations* (ICLR) Domain Generalization Workshop, 2023. [Link]

- [9] J. Park, K. Kaai, **S. Hossain**, N. Sumi, S. Rambhatla, P. Fieguth. Building Spatio-temporal Transformers for Egocentric 3D Pose Estimation. Joint International Workshop on Egocentric Perception, Interaction and Computing (EPIC) and Ego4D, *IEEE/CVF Computer Vision and Pattern Recognition Conference* (CVPR), 2022. [Link]
  ⋆ *Oral presentation.*

Experience

Machine Learning Engineer May 2025 – Present
Deep Breathe *London, Ontario, Canada*

- Spearheading development of multiple ML models for diagnostic ultrasound that run on-device in real time.
- Set up Slurm on the team's GPU server and ran 500+ training experiments across it and GCP Vertex AI.
- Built and led the labeling infrastructure on V7 Darwin (GCP, SQL) from scratch, used to label millions of frames.
- Automated clip ingestion into the training database, labeling-sprint setup and pre-label uploads through the V7 Darwin API.
- Built a PyTorch tracker propagating Labelbox annotations across frames, cutting labeling time per video 5x (10 to 2 min).

Machine Learning Research Intern Jan 2024 – Aug 2024
Kolena *San Francisco, California, USA*

- Fine-tuned an LLM judge to rate Qwen-VL descriptions against human preferences, cutting manual review volume 3x.
- Raised RAG answer accuracy on financial PDFs from 80% to 93% by rebuilding parsing and prompting (LangChain).
- Built a Streamlit + OpenAI SDK tool that drafts Q&A and similar test samples, making text-dataset creation 5x faster.

- Shipped CV and NLP metadata extraction (OpenCV, NLTK, spaCy) in Kolena's AI Quality Platform so users can stratify test sets and find model errors on edge cases; presented it in a webinar.

Researcher Sep 2023 – Present
Critical ML Lab *Waterloo, Ontario, Canada*

- First-authored TamperBench with FAR.AI (**KDD '26 Oral, top 4.5%**): stress-tested 21 open-weight LLMs against 9 tampering attacks in 10K+ LoRA/full fine-tuning runs (20K GPU-hours, up to 8 GPUs); safeguards fell on all 21.
- Set TamperBench's initial roadmap and task split for an 11-author team from 5 institutions.
- Quoted in MIT Technology Review and University of Waterloo News (2026) as TamperBench's lead author.
- Created SafeTuneBed (IJCAI '25 workshop), an open-source toolkit benchmarking LLM safety defenses under fine-tuning.
- Built FOND's training/eval pipelines (T-PAMI '25); the method lifted tumor-detection accuracy 9% at unseen hospitals.
- Co-authored Apple-funded LangDA: state-of-the-art domain-adaptive semantic segmentation with vision-language models (NeurIPS '24 workshop; Best Vision Paper, CVIS '24).

Machine Learning Developer (Co-op & Part-time) Sep 2022 – Apr 2025
Deep Breathe *London, Ontario, Canada*

- Led development of an effusion classifier: 0.90 sensitivity, 0.89 specificity on a 239-clip holdout from 2 hospitals (AJEM '25).
- Converted an earlier lung-ultrasound model for on-device inference in Deep Breathe's mobile apps.
- Built Azure/SQL data pipelines (Python, Bash, OpenCV) that ingest, process and version ultrasound data, generating 7.8M M-mode images from 140K B-mode videos per version.
- Cut 1K-clip dataset builds from 1–2 hours to minutes: teams now download prebuilt, versioned datasets.
- Integrated Weights & Biases into ML projects for training dashboards and evaluation logging.

Software Engineering Intern Jan 2022 – Apr 2022
IntelliSports *Montreal, Quebec, Canada*

- More than doubled IMU cheat-detection sensitivity (25% to 55%) via feature engineering with scikit-learn and SciPy.
- Deployed punch, crunch, high-knee and jump detection on AWS EC2, growing the app's workout missions from 2 to 5.
- Sped up CPU inference 1.8x by switching to lightweight scikit-learn models.
- Found, via decision-tree analysis of early user data, an onboarding path whose users were 1.5x likelier to be retained.
- Built a PHP/MySQL analytics dashboard of user activity growth, used in pitches that secured $1.4M in funding.

Undergraduate Research Assistant May 2021 – Aug 2023
University of Waterloo *Waterloo, Ontario, Canada*

- Co-developed Nissan-funded Ego-STAN, spatio-temporal Transformers for egocentric 3D pose: 38% lower error on hardest joints with 22% fewer parameters (KDD '23; CVPR '22 workshop oral).
- Co-authored Implicit Stylization for Domain Adaptation (ICLR '23 Domain Generalization workshop).
- Implemented IMU feature extraction for human-robot interaction (Social and Intelligent Robotics Research Lab, 2021).
- Implemented face identification (dlib, 99.3% on LFW) and emotion recognition (72.7% on FER2013) on the REEM-C humanoid robot with ROS, Flask and PostgreSQL (HCRMI Lab co-op, 2021).
- Designed a multi-input Keras model for facial emotion recognition (pybind11 bindings of a C++ toolkit; t-SNE analysis).

Honours & Awards

- Alumni Gold Medal – *Finalist*
- Best Capstone Design Project – Biomedical Engineering *$500*
- CRA Outstanding Undergraduate Researcher Award – *Honorable Mention*
- Best Vision Paper Award *Conference on Vision and Intelligent Systems* 2024
- Yuen Family Foundation Award for Healthy Aging – Final-Year Design Project *$10,000*
- President's Research Award *$1,500*
- Engineering International Student Scholarship *$10,000*
- President's Scholarship of Distinction *$2,000*
- Term Dean's Honours List: Fall 2023, Winter 2023, Fall 2021, Fall 2020
- Top AS Level Exam Scorer in Country *Cambridge International Examinations* 2019
- Co-op Performance – *Outstanding* at Kolena (Winter & Spring 2024), Deep Breathe (Fall 2022 & Spring 2023), IntelliSports (Winter 2022), HCRMI Lab (Spring 2021)

Panels and Media

- Open models have real benefits. Do they outweigh the risks? MIT Technology Review. *Quoted*, 2026. [Link]
- Major security weaknesses found in leading open AI models. University of Waterloo News. *Quoted*, 2026. [Link]
- Tests raise fears open AI models could be used at scale for harm. Waterloo Engineering News. *Research Featured*, 2026. [Link]
- LLM Safety Under Fine-Tuning. AIWILD, CAO and RSI Workshops at ICLR. *Presenter*, 2026.
- Translational Pathways in Biomedical Engineering: From Lab to Clinic. CUBEC. *Panelist*, 2026. [Link]
- BME undergraduate student received Honorable Mention for Outstanding Undergraduate Researcher Award. University of Waterloo Systems Design Engineering News. *Featured*, 2025. [Link]
- Augmented Reality for Guided Lung Ultrasound Acquisition. Ibtikar Seminar Series. *Guest Speaker*, 2025. [Link]
- Enhancing AI Testing: The Power of Metadata Hydration. Kolena Webinar. *Key Speaker*, 2024. [Link]
- Mastering RAG Systems for LLMs: From Pitfalls to Performance. Kolena Webinar. *Panelist and Slide Author*, 2024. [Link]
- IMU Feature Extraction. Physiological Computing Hands-On Workshop, University of Waterloo. *Presenter*, 2022. [Link]

Projects

- PneumoGuide: Augmented Reality to Guide Lung Ultrasound Acquisition. *Final-Year Design Project*, 2025. Best Capstone Design Project; $10,000 Yuen Family Foundation Award for Healthy Aging.
- Fareeka: An Audiobook Platform for Classical Arabic Texts. | *Vercel, PostgreSQL, OpenAI, ElevenLabs*
- Investigating Safety Alignment Properties of Parameter-Efficient Fine-Tuning Methods. *MSE 598: Applied Engineering Research.* | *PyTorch, Hugging Face, GaLore*
- Stylization for Label-Scarce Unsupervised Domain-Adaptive Semantic Segmentation. *SYDE 577: Deep Learning.* | *PyTorch*
- HRI Physio Lib: Physiological Adaptation in HRI. *Contributed IMU Signal Processing Tools*, 2022. [Link] [GitHub]
- Brain-Wave Gaming & Speech. *Demo post with 60K+ impressions.* [GitHub] | *Python, OpenAI, SciPy*

Technical Skills

Languages: Python, C++, C#, SQL, Bash, PHP, HTML
ML: PyTorch, TensorFlow, Keras, Hugging Face Transformers, PEFT/LoRA, vLLM, LangChain, OpenAI SDK, scikit-learn, SciPy, NumPy, pandas, OpenCV, dlib, NLTK, spaCy, Weights & Biases
Infrastructure: Linux, Git, Docker, Slurm, GCP (Vertex AI), AWS (EC2), Azure, multi-GPU training, ROS, Flask, Streamlit
Data and labeling: MySQL, PostgreSQL, V7 Darwin, Labelbox
Areas: deep learning, computer vision, medical imaging, large language model (LLM) fine-tuning & safety evaluation, RAG, domain adaptation, pose estimation, human-robot interaction